DATA ANALYTICS PROJECT

UK Train Rides

Turning railway data into better journeys

A data-driven look at ticket sales, routes, passenger demand and service performance.

Smarter insights • Better decisions • Improved passenger experience

TEAM LEADER | SHADWA AHMED

01 / PROJECT OVERVIEW

Understanding UK Train Rides

A focused analytics proposal for exploring rail journeys and operational patterns.

Primary objective

Transform train-ride data into clear, interactive insights that help explain travel demand, ticket revenue and service performance.

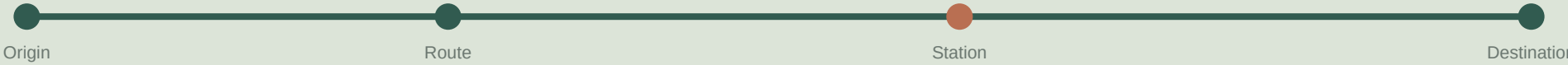
Project scope

Explore journeys, stations, ticket types, rail routes, travel dates and available delay or cancellation information in the dataset.

Business value

Support evidence-based decisions around busy routes, passenger demand, ticket sales and opportunities to improve the travel experience.

From raw records to useful decisions



02 / ANALYTICAL FOCUS

Questions the Analysis Will Answer

The final questions and KPIs will be aligned with the columns available in the project dataset.

01 Passenger demand

Which routes, stations and time periods show the highest travel activity?

02 Ticket sales & revenue

How do ticket types, classes and purchase patterns relate to sales and revenue?

03 Journey performance

Where are delays or cancellations more frequent, if those fields are available?

04 Route comparison

How do journey volume and revenue vary across stations and destinations?

03 / TECHNICAL DELIVERABLE

Interactive Analytics Dashboard

A clear, filterable view of the patterns hidden in train-ride data.

UK Train Rides | Overview

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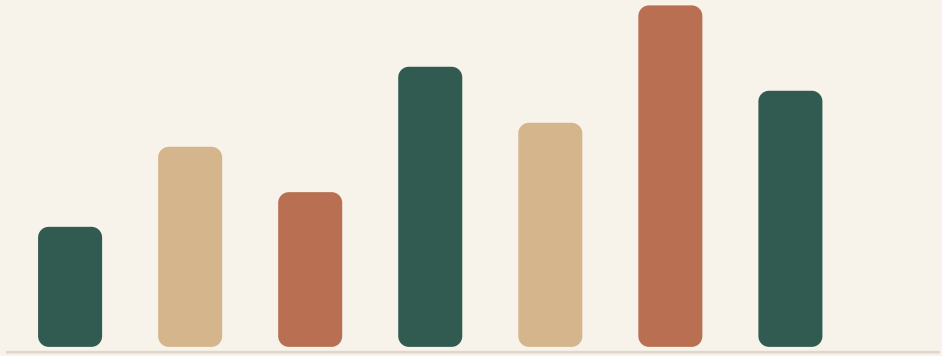
Total journeys	Ticket revenue	Delay rate*

Illustrative visual layout — final values come from the dataset



Dashboard features

- **Dynamic filters**
  Date, route, station, ticket type and other available fields.
- **Visual KPIs**
  Journey volume, revenue and service indicators supported by the data.
- **Comparisons**
  Rank routes and stations; compare patterns across time.
- **Clear reporting**
  A compact summary designed for quick review and discussion.

*Delay-related metrics will be included only if supported by the dataset.

04 / IMPLEMENTATION PLAN

Four-Week Project Roadmap

A practical workflow from data understanding to a polished presentation.

WEEK 1

Data understanding

Review columns, data types, missing values and data quality.

WEEK 2

Data preparation

Clean records, standardize fields and build a suitable data model.

WEEK 3

Analysis & visuals

Create measures, compare patterns and build dashboard pages.

WEEK 4

Validation & delivery

Check findings, refine design and prepare the final presentation.

Working principle

Every chart and metric should answer a clear question and be validated against the source data.

05 / EXPECTED OUTCOMES

What the Project Can Deliver

The aim is to reveal evidence-based opportunities, not to assume results before analysis.

Clearer demand patterns

Identify busier routes, stations and periods to support planning.

Better sales visibility

Understand how ticket types and journey patterns relate to revenue.

Performance monitoring

Highlight delay or cancellation patterns where the source data supports them.

Faster decisions

Bring key findings into one consistent dashboard instead of scattered tables.

Note: any quantified impact will be reported only when supported by analysis.

06 / TOOLS & TEAM

How We Will Work Together

A collaborative approach with clear responsibilities and a shared final output.

Power BI

Data modeling, measures, interactive visuals and dashboard design.

Excel / Power Query

Initial data inspection, cleaning, transformation and validation.

Project documentation

Record assumptions, definitions, findings and final recommendations.

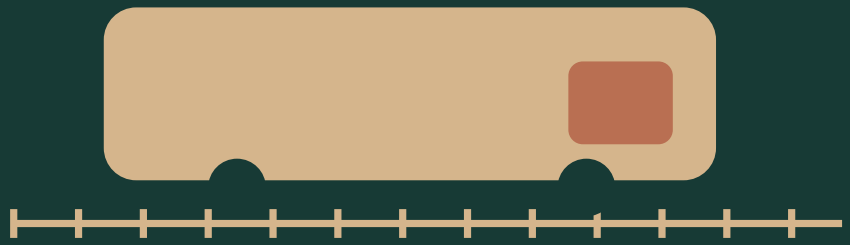
Project team

- **Shadwa Ahmed** — Team Leader
- **Hager Ahmed** — Team Member
- **Youmna Hussein** — Team Member
- **Mohamed Tarek** — Team Member
- **Omar Adel** — Team Member
- **Abdelrahman Ayman** — Team Member

UK TRAIN RIDES • DATA ANALYTICS

Better journeys start with better insights.

Explore the data. Understand the patterns. Support smarter decisions.



Thank you

Questions & discussion

TEAM LEADER: SHADWA AHMED